

First Article Inspection Procedure

Purpose

This work instruction defines the procedure for performing First Article Inspection (FAI) on new or modified products at Acme Corporation.

Scope

This instruction applies to all first article inspections required for new products, design changes, process changes, and supplier changes.

Definitions

First Article: The first production part produced under new or modified conditions.

FAI Report: A comprehensive document recording the results of the first article inspection.

Dimensional Report: A record of all measured dimensions compared to the specification.

When FAI is Required

FAI is required in the following situations:

1. New product introduction
2. Design change affecting form, fit, or function
3. Process change (new tooling, equipment, or method)
4. Material change
5. Supplier change for a critical component
6. Production relocation to a different line or facility
7. After a production shutdown of more than 12 months

Procedure

Step 1: Preparation

1. Obtain the product specification, engineering drawing, and bill of materials.
2. Review the critical characteristics and special requirements.
3. Prepare the inspection plan including:
 - All dimensions and tolerances from the drawing
 - Material and finish requirements
 - Functional test requirements
 - Visual acceptance criteria per WI-042
4. Verify all measurement equipment is calibrated per WI-019.
5. Obtain the first production part or sample.

Step 2: Dimensional Inspection

1. Measure every dimension specified on the engineering drawing.
2. Record each measurement value on the dimensional report.
3. Compare each measurement to the specified tolerance.
4. Mark each dimension as pass or fail.
5. If any dimension fails, notify the QA Engineer immediately.

Step 3: Material and Finish Verification

1. Verify material composition against the specification.
2. Verify surface finish against the specified requirement.
3. Verify any coatings, platings, or treatments.
4. Record results on the FAI report.

Step 4: Functional Testing

If the product specification requires functional testing:

1. Set up the test according to the test procedure.
2. Perform the test and record results.
3. Verify that results meet acceptance criteria.
4. Document any deviations.

Step 5: FAI Report Completion

The FAI report must include:

- Part number and revision level
- Drawing number and revision
- Supplier name and part number (if applicable)
- Date of inspection
- Dimensional report with all measurements
- Material and finish verification results
- Functional test results (if applicable)
- List of non-conformances (if any)
- Inspector name and signature
- QA Engineer review and approval

Step 6: Disposition

Condition	Disposition
All dimensions and requirements pass	FAI approved. Production may start.
Minor non-conformance (form, fit, or function)	FAI conditionally approved with deviation permit.

Condition	Disposition
Major non-conformance	FAI rejected. Root cause analysis required. Re-inspect after correction.

Step 7: Record Retention

FAI reports are retained for the life of the product plus 5 years. Records are maintained in the QA document control system.

Related Documents

- WI-042: Visual Inspection Criteria for Finished Products
- WI-019: Calibration of Measurement Equipment
- SOP-QA-015: Supplier Quality Management

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